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Course: Calculus Beta

Assignment: MyLab Math - Uge 2

Use l'Hopital's Rule to evaluate $\lim_{t \rightarrow 0} \frac{6 \sin(4t^3)}{-5t}$.

First, find the limits of the numerator and the denominator.

$$\lim_{t \rightarrow 0} (-5t) = 0, \quad \lim_{t \rightarrow 0} (6 \sin(4t^3)) = 0$$

Evaluating the limit, $\lim_{t \rightarrow 0} \frac{6 \sin(4t^3)}{-5t}$, by direct substitution results in the indeterminate form $\frac{0}{0}$.

L'Hôpital's Rule states that if $f(a) = g(a) = 0$, f and g are differentiable on an open interval I containing a , and $g'(x) \neq 0$ on I if $x \neq a$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \text{ assuming that the limit on the right exists.}$$

Apply l'Hôpital's Rule to $\lim_{t \rightarrow 0} \frac{6 \sin(4t^3)}{-5t}$.

$$\lim_{t \rightarrow 0} \frac{6 \sin(4t^3)}{-5t} = \lim_{t \rightarrow 0} \frac{72t^2 \cos(4t^3)}{-5}$$

Evaluate the limit using the derivatives.

$$\lim_{t \rightarrow 0} \frac{72t^2 \cos(4t^3)}{-5} = 0$$

$$\text{So, } \lim_{t \rightarrow 0} \frac{6 \sin(4t^3)}{-5t} = \lim_{t \rightarrow 0} \frac{72t^2 \cos(4t^3)}{-5} = 0.$$